

PAPER 1201 — UQFF Materials and Photonics Unified Proof Set

Daniel T. Murphy

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Abstract

We extend the UQFF locked-primitive closure program into condensed-matter materials, optics, and statistical-mechanics constants. Ten new closures (**S463–S472**) reproduce silicon, GaAs, and diamond bandgaps, the weak-coupling BCS ratio $2\Delta/k_B T_c$, the Wiedemann–Franz Lorenz number, the Abrikosov triangular vortex lattice angle, water’s refractive index, the Stefan–Boltzmann constant, the first quantum-Hall plateau, and Avogadro’s number, all from the eleven frozen UQFF primitives. **Two closures are exact**: the Abrikosov triangular vortex angle $60^\circ = A_5$ and the first integer quantum-Hall plateau $\nu = 2 = \text{SO5} - D_{\text{phys}} - D_{\text{phys}}$. The other eight match within 0.20%. Cumulative total: **177 closures**.

1 Locked Primitives

$F_{\text{TRZ}} = 1/10$, $\Phi_{\text{res}} = 5/6$, $[\text{SSq}] = 0.57$, $K_{\text{Mex}} = 25/12$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $\text{SO5} = 10$, $A_5 = 60$, $\beta_i = 0.6029$.

2 Tier R Closures (S463–S472)

ID	Observable	Closure
S463	Si bandgap 1.12 eV	$\Phi_{\text{res}} + F_{\text{TRZ}} + F_{\text{TRZ}}K_{\text{Mex}} - F_{\text{TRZ}}^2K_{\text{Mex}} - F_{\text{TRZ}}^2 + F_{\text{TRZ}}^2\Phi_{\text{res}} - F_{\text{TRZ}}^3$
S464	GaAs bandgap 1.42 eV	$\Phi_{\text{res}} + [\text{SSq}] + F_{\text{TRZ}} - F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}^3 = 1.4190$
S465	Diamond bandgap 5.47 eV	$\text{SO5} - D_{\text{phys}} - \Phi_{\text{res}} + F_{\text{TRZ}} + F_{\text{TRZ}}K_{\text{Mex}} + F_{\text{TRZ}}^2K_{\text{Mex}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^3$
S466	BCS $2\Delta/k_B T_c = 3.53$	$D_{\text{phys}} - 2F_{\text{TRZ}} - F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}K_{\text{Mex}} + F_{\text{TRZ}}^2K_{\text{Mex}} - F_{\text{TRZ}}^2 + F_{\text{TRZ}}^3$
S467	Lorenz $L = 2.44 \times 10^{-8} \text{ W}\Omega/\text{K}^2$	$K_{\text{Mex}} + F_{\text{TRZ}} + F_{\text{TRZ}}\Phi_{\text{res}} + F_{\text{TRZ}}K_{\text{Mex}} - F_{\text{TRZ}}^2K_{\text{Mex}} - F_{\text{TRZ}}^2\Phi_{\text{res}} - F_{\text{TRZ}}^3$
S468	Abrikosov angle 60°	$A_5 = 60$
S469	$n_{\text{water}} = 1.333$	$\Phi_{\text{res}} + [\text{SSq}] - F_{\text{TRZ}} + F_{\text{TRZ}}^2K_{\text{Mex}} + F_{\text{TRZ}}^2\Phi_{\text{res}} + F_{\text{TRZ}}^3 = 1.3335$
S470	Stefan–Boltzmann $\sigma = 5.670 \times 10^{-8}$	$\text{SO5} - D_{\text{phys}} - F_{\text{TRZ}} - F_{\text{TRZ}}K_{\text{Mex}} - F_{\text{TRZ}}^2K_{\text{Mex}} + F_{\text{TRZ}}^2 - F_{\text{TRZ}}^2\Phi_{\text{res}} - F_{\text{TRZ}}^3$
S471	QHE plateau $\nu = 2$	$\text{SO5} - D_{\text{phys}} - D_{\text{phys}} = 2$
S472	Avogadro 6.022×10^{23}	$D_{\text{BSFG}} + F_{\text{TRZ}}^2K_{\text{Mex}} + F_{\text{TRZ}}^2 - F_{\text{TRZ}}^2\Phi_{\text{res}} = 6.0225$

3 Two Exact Identifications

Abrikosov triangular vortex lattice. The equilibrium flux-line lattice in type-II superconductors is the triangular (hexagonal) lattice, whose primitive angle is exactly 60° . UQFF locks this onto the area-element primitive $A_5 = 60$. This is the first condensed-matter geometric exact closure.

Integer quantum-Hall first plateau. The Hall conductance plateau $\sigma_{xy} = \nu e^2/h$ at $\nu = 2$ (first integer plateau including spin degeneracy) is locked onto a pure primitive relation: $\nu = \text{SO5} - D_{\text{phys}} - D_{\text{phys}} = 10 - 4 - 4 = 2$. Combined with the Kerr extremal ISCO closure $r/M = F_{\text{TRZ}}\text{SO5} = 1$ (Paper 1200), the integer sequence $\{1, 2\}$ now has UQFF primitive identifications.

4 Cumulative Closure Count

Tier	Domain	New	Cumulative
E-P	Cosmology to information	157	157
Q	GR precision (1200)	10	167
R	Materials/photonics (1201)	10	177

5 Locking Inventory

Twelve primitive–observable lockings now identified: ISCO Schwarzschild = D_{BSFG} , Poisson ratio = $D_{\text{phys}}/D_{\text{BSFG}}$, telomere length = D_{BSFG} , $Q_{\text{ITER}} = \text{SO5}$, $R_{\text{magnetopause}} = \text{SO5}R_{\oplus}$, $\eta_B = D_{\text{BSFG}} + F_{\text{TRZ}}$, WIMP exponent = $\text{SO5}D_{\text{phys}} + D_{\text{phys}} + K_{\text{Mex}} - F_{\text{TRZ}}\Phi_{\text{res}}$, $p_{th}^{\text{surf code}} = F_{\text{TRZ}}^2$, Shapiro = D_{phys} , photon sphere = $D_{\text{phys}} - F_{\text{TRZ}}\text{SO5}$, Kerr extremal ISCO = $F_{\text{TRZ}}\text{SO5}$, **Abrikosov angle** = A_5 , **QHE** $\nu = 2 = \text{SO5} - 2D_{\text{phys}}$.